

Validation of Yield Curve Construction in M-Library

ver. 1.0

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1. Introduction

1.1. Purpose and Scope

This document reports the validation results on the yield curves built in M-Library. M-Library supports a full variety of curves including OIS Curve, Swap Curve, Tenor Basis Curve, Cross Currency Basis Curve (USD CSA Curve), FX Forward Constant Curve (Non-USD CSA Curve) and Cheapest-to-Deliver Curves. The implementation is analytical, and the optimization is done with the Newton-Raphson algorithm.

In Scope

- ✓ The curves in scope are OIS, Swap, Tenor Basis and Cross Currency Basis Curves (USD CSA Curve).
- ✓ The currencies in scope are USD, GBP, EUR, AUD and JPY.
- ✓ The validation is mainly done through repricing of all market quotes.
- ✓ The smoothness of forward curves is visually checked.
- ✓ The benchmark yield curves are constructed by the iterative bootstrap.

Out of Scope

- ✓ FX Forward Constant Curves (Non-USD CSA curves) and Cheapest-to-Deliver Curves are not validated.
- ✓ When introducing Libor-OIS Basis Spread, M-Library constructs the synthetic OIS from Libor-OIS Basis Swap and Swap in order to avoid circular dependency. Any global solving approach to solve OIS and Swap Curves simultaneously has not tried in this document.

1.2. Summary of Validation Results

- ✓ M-Library correctly repriced all the market quotes at par, with accuracy greater than or equal to $1/10,000^{\text{th}}$ of a basis point for almost all of the instruments. The only instrument to be noted is that AUD 30Y is the order of $1/1000^{\text{th}}$ basis point, but we suppose it is not a problem, as it is in the bid-offer spread. (More details in 3.2.)
- ✓ M-Library zero curves are brought into our risk management department (hereafter RMD) library, and independently confirmed those zero curves could reprice the market quotes at par with enough accuracy. This suggests that the M-Library's implementation is generally accurate. (More details in 3.3.)
- ✓ The discrepancy in repricing was observed when M-Library and RMD library used different cashflow schedule. M-Library does not hold fixing date information. For interest rate, M-Library uses the accrual start date instead of fixing date + spot lag, so the actual differences between the accrual start date and fixing date + spot lag will be ignored. And for FX, M-Library uses the payment date instead of fixing date + spot lag, so again, the actual differences between the payment date and fixing date +

spot lag will be ignored. The differences are occasionally caused by holiday adjustments or differences between trades' specific fixing calendars and market common conventions of the instruments. (More details in 3.3.)

- ✓ The forward curves are generally smooth, except for OIS curves. It is especially worth mentioning that the Swap curves remain stable during futures contract roles. (More details in 4.1.)
- ✓ The forward OIS curves in M-Library tend to have zigzag shape at short end and curtain-like shape at long end. However we think these shapes are not crucial in terms of risk management, because we confirmed Overnight Index Swap were repriced correctly, and other than Overnight Index Swap, the OIS curve is used for discounting only. (More details in 4.1.)
- ✓ The M-Library curves generally match with our benchmark curves independently built in RMD library. Our benchmark curves are constructed by the iterative bootstrap. The forward OIS curves from RMD library look smoother, and the curtain-like shape at long end has disappeared. (More details in 4.2. and 4.3)

1.3. Conclusion

M-Library correctly repriced all the market quotes at par, with accuracy greater than or equal to 1/10,000th of a basis point for almost all of the instruments. The only instrument to be noted is that AUD 30Y Swap is the order of 1/1000th basis point, but we suppose it is not a problem, as it is in the bid-offer spread.

M-Library may be able to introduce the fixing date information of interest rate and FX for more accurate curve building, but it is currently not necessary considering the repricing accuracy.

For the yield curve shape, the forward curves of M-Library are generally smooth, except for OIS curves. It is especially worth mentioning that the Swap curves remain stable during futures contract roles. The forward OIS curves in M-Library tend to have zigzag shape at short end and curtain-like shape at long end. However we think it is not crucial in terms of risk management, because we confirmed Overnight Index Swap were repriced correctly, and other than Overnight Index Swap, the OIS curve is used for discounting only.

We also understand that Quant research & Analytics in Mizuho International plc are currently extending their curves to resolve the OIS curve shape issues and adding dual-bootstrapping of OIS and standard Swap curves.

1.4. M-Library Add-in Version

The version number of M-Library Add-in validated in this document is 20170915.2221. It was last built on September 15th, 2017 by Quant research & Analytics in Mizuho International plc.

2. Curve Construction in M-Library

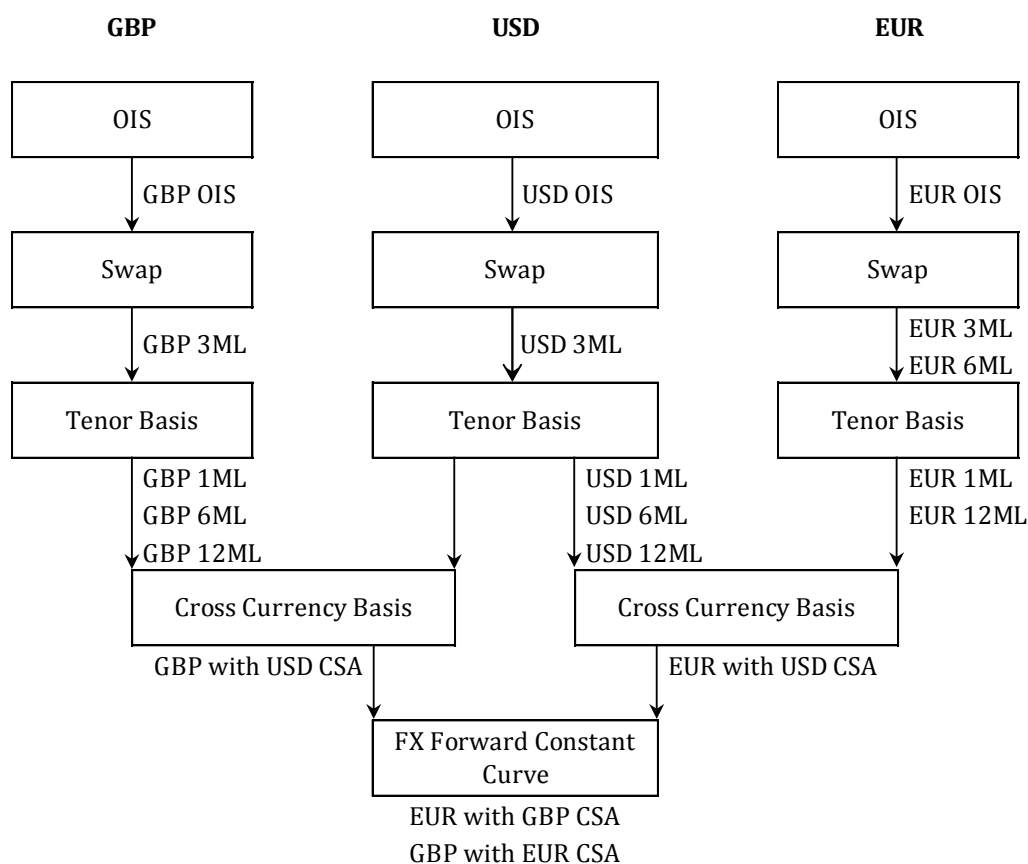
2.1. Curve Construction

Curve construction and Currency Dependencies

M-Library supports a full variety of curves including:

- ✓ OIS Curve
- ✓ Swap Curve
- ✓ Tenor Basis Curve
- ✓ Cross Currency Basis Curve (USD CSA Curve)
- ✓ FX Forward Constant Curve (Non-USD CSA Curve)
- ✓ Cheapest-to-Deliver Curves

The below chart illustrates how those curves are constructed by taking an example of USD, GBP and EUR. OIS, Swap and Tenor Basis Curves can be constructed for individual currencies, but USD CSA Curves and Non-USD CSA Curves require additional currency dependencies. As M-Library assumes FX Forwards are constant and fixed assuming USD CSA collateral, USD is placed in the center. More detailed will follow in 2.5 and 2.6.



Interpolation Scheme and State Variables

Interpolation schemes prepared in M-Library are:

- ✓ fn_StepInterpolation Piecewise Constant
- ✓ fn_LinearInterpolation Linear
- ✓ fn_MonotoneConvexInterpolation Monotone Convex
- ✓ fn_SplineInterpolation Spline
- ✓ fn_ConstrainedSplineInterpolation Constrained Spline
- ✓ fn_LinearSplineInterpolation Linear (for Deposits, FRA/Future)+ Spline (for Swap)

The default settings of interpolation schemes are given by Quant research & Analytics in Mizuho International plc as below. These settings are same for all currencies. The state variables are automatically selected within the M-library, and the users can't select them on their own choice.

	Interpolation	State Variables
OIS Curves	Spline	Zero rates
Swap Curves	Linear+Spline	Deposit Rates, FRAs/Futures: Discount factors Swaps: Logarithm of discount factors
Basis Curves (Tenor Basis and Xccy Basis)	Spline	FX Forwards: Discount factors Basis Swaps: Logarithm of target curve's discount factors, or, logarithm of spreads added on top of the base curve

Table 2.1. Interpolation and State Variables in M-Library

2.2. OIS Curves

The OIS Curve is calibrated from OIS outright, Central Bank Swaps if any and Libor-OIS Spreads.

- ✓ USD OIS outright out to 2Y inclusive, Libor-OIS Basis Swaps from 3Y inclusive
- ✓ GBP OIS outright out to 2Y inclusive, Libor-OIS Basis Swaps from 3Y inclusive
- ✓ EUR OIS outright out to 2Y inclusive, Libor-OIS Basis Swaps from 3Y inclusive
Central Bank Swaps (ECB)
- ✓ AUD OIS outright only
- ✓ JPY OIS outright out to 2Y inclusive, Libor-OIS Basis Swaps from 3Y inclusive

Overnight Index Swap

With the market quote of OIS, R^{OIS} , Overnight Index Swap is defined with the following equation:

$$R^{OIS} \sum_{i_1=0}^{n_1-1} \tau(T_{i_1}, T_{i_1+1}) P^{OIS}(0, T_{i_1+1}) = \sum_{i_2=0}^{n_2-1} F^{OIS}(T_{i_2}, T_{i_2+1}) \tau(T_{i_2}, T_{i_2+1}) P^{OIS}(0, T_{i_2+1}).$$

Denoting the over-night time interval, $T_{i_2} = T_{i_2,0} < \dots < T_{i_2,m} = T_{i_2+1}$, the OIS rate is estimates as the geometric compounded rate for EUR, AUD and JPY:

$$F^{OIS}(T_{i_2}, T_{i_2+1}) = \frac{\prod_{j=0}^{m-1} (1 + F_{on}^{OIS}(T_{i_2,j}, T_{i_2,j+1}) \tau(T_{i_2,j}, T_{i_2,j+1})) - 1}{\tau(T_{i_2}, T_{i_2+1})},$$

and is estimated as the arithmetic average rate for USD and GBP:

$$F^{OIS}(T_{i_2}, T_{i_2+1}) = \frac{\sum_{j=0}^{m-1} F_{on}^{OIS}(T_{i_2,j}, T_{i_2,j+1}) \tau(T_{i_2,j}, T_{i_2,j+1})}{\tau(T_{i_2}, T_{i_2+1})}.$$

$F_{on}^{OIS}(T_{i_2,j}, T_{i_2,j+1})$ is the single overnight rate spanning the time interval $(T_{i_2,j}, T_{i_2,j+1}]$ and estimated as:

$$F_{on}^{OIS}(T_{i_2,j}, T_{i_2,j+1}) = \frac{1}{\tau(T_{i_2,j}, T_{i_2,j+1})} \left\{ \frac{P^{OIS}(0, T_{i_2,j})}{P^{OIS}(0, T_{i_2,j+1})} - 1 \right\}.$$

Pillar date of Overnight Index Swap is set to be the maturity tenor of a given instrument. The discount factors of all pillar dates are sequentially solved by Newton Raphson algorithm.

Central Bank Swap

In essence, during the course of each Central Bank Swap, daily discount factors are derived assuming the *constant overnight forward rate* $F(t, T)$ spanning the time interval $(t, T]$, $t = t_0 < \dots < t_m = T$, that is equal to the Central Bank Swap quote:

$$P^{OIS}(0, t_{i+1}) = P^{OIS}(0, t_i) \frac{1}{1 + F(t, T) \tau(t_i, t_{i+1})}.$$

Only if there is a gap between the maturity of last OIS and the start of the first Central Bank Swap, the *constant overnight forward rate* is linearly interpolated between the last OIS rate and the first Central Bank Swap rate, and the discount factor on the start of first Central Bank Swap is estimated from there.

Libor-OIS Swap

With the market quote of Libor-OIS Basis Spread, $R_{basis}^{Libor-OIS}$, Libor-OIS Swap is defined with the following equation:

$$\begin{aligned} & \sum_{i_1=0}^{n_1-1} (F^{OIS}(T_{i_1}, T_{i_1+1}) + R_{basis}^{Libor-OIS}) \tau(T_{i_1}, T_{i_1+1}) P^{OIS}(0, T_{i_1+1}) \\ &= \sum_{i_2=0}^{n_2-1} F^{Libor}(T_{i_2}, T_{i_2+1}) \tau(T_{i_2}, T_{i_2+1}) P^{OIS}(0, T_{i_2+1}). \end{aligned}$$

The pricing of Libor-OIS Swap requires both OIS curve and Libor Curve, as the forward Libor rate is estimated as

$$F^{Libor}(T_{i_2}, T_{i_2+1}) = \frac{1}{\tau(T_{i_2}, T_{i_2+1})} \left\{ \frac{P^{Libor}(0, T_{i_2})}{P^{Libor}(0, T_{i_2+1})} - 1 \right\}.$$

To avoid the circular dependency, M-Library introduce the synthetic OIS by combining the Libor-OIS Swap and the standard Libor Swap. Denoting R^{swap} the market quote of Swap, the Libor Swap is defined as:

$$R^{swap} \sum_{i_3=0}^{n_3-1} \tau(T_{i_3}, T_{i_3+1}) P^{OIS}(0, T_{i_3+1}) = \sum_{i_2=0}^{n_2-1} F^{Libor}(T_{i_2}, T_{i_2+1}) \tau(T_{i_2}, T_{i_2+1}) P^{OIS}(0, T_{i_2+1}).$$

Therefore the synthetic OIS is led as:

$$R^{swap} \sum_{i_3=0}^{n_3-1} \tau(T_{i_3}, T_{i_3+1}) P^{OIS}(0, T_{i_3+1}) = \sum_{i_1=0}^{n_1-1} (F^{OIS}(T_{i_1}, T_{i_1+1}) + R_{basis}^{Libor-OIS}) \tau(T_{i_1}, T_{i_1+1}) P^{OIS}(0, T_{i_1+1}).$$

Pillar date of Libor-OIS Swap is set to be the maturity tenor of a given instruments. The discount factors of all pillar dates are sequentially solved by one-dimensional Newton Raphson algorithm.

2.3. Swap Curves

The Swap Curve is calibrated from Deposit Rates, Futures/FRA's and Swaps.

- ✓ USD Libor fixings, Futures, Swaps
- ✓ GBP Libor fixings, Futures, Swaps
- ✓ EUR Libor fixings, Futures/FRA's, Swaps
- ✓ AUD Libor fixings, FRA's, Swaps
- ✓ JPY Libor fixings, FRA's, Swaps

M-Library treats Swap section differently from Deposit Rate and Future/FRA section. The Deposit

Rate/Future/FRA interpolator carries the discount factors, whereas the Swap interpolator carries the logarithm of discount factors. Also, the Deposit Rate/Future/FRA section finds the discount factors sequentially, whereas the Swap section tries to solve the discount factors of all maturity tenors simultaneously by using the multivariate Newton Raphson algorithm.

Deposit Rate

With the market quote of Deposit Rate such as Libor fixing, $F(0, T)$, the discount factor for a certain tenor is derived as:

$$P^{\text{Libor}}(0, T) = \frac{1}{1 + F(0, T)\tau(0, T)}.$$

FRA

With the market quote of FRA, $F^{\text{FRA}}(t, T)$, the discount factor on the maturity tenor of a given FRA instrument is led as:

$$P^{\text{Libor}}(0, T) = P^{\text{Libor}}(0, t) \frac{1}{1 + F^{\text{FRA}}(t, T)\tau(t, T)}.$$

$P^{\text{Libor}}(0, t)$ is interpolated from the discount factors obtained from the calibration instruments prior to the current FRA.

Future

With the market quote of Future price, $P^{\text{future}}(t, T)$, the forward rate is derived as:

$$F^{\text{future}}(t, T) = 1 - P^{\text{future}}(t, T)/100.$$

As M-Library supports the convexity adjustment by 1 factor Hull White model, the discount factor is led with consideration of convexity adjustment as:

$$F^{\text{future}}(t, T) = \frac{1}{\tau(t, T)} \left\{ \frac{P^{\text{Libor}}(0, t)}{P^{\text{Libor}}(0, T)} - 1 \right\} + \frac{B(t, T)}{T - t} (B(t, T)(1 - e^{-2at}) + 2aB(0, t)^2) \frac{\sigma^2}{4a},$$

where

$$B(t, T) = \frac{1 - e^{-a(T-t)}}{a}.$$

Swap

With the market quote of Swap Rate, R^{swap} , Swap is defined with the following equation:

$$R^{\text{swap}} \sum_{i_1=0}^{n_1-1} \tau(T_{i_1}, T_{i_1+1}) P^{OIS}(0, T_{i_1+1}) = \sum_{i_2=0}^{n_2-1} F^{\text{Libor}}(T_{i_2}, T_{i_2+1}) \tau(T_{i_2}, T_{i_2+1}) P^{OIS}(0, T_{i_2+1}).$$

The Swap section of Swap Curve builder tries to solve the discount factors of all Swap maturity tenors simultaneously by using the multivariate Newton Raphson algorithm. Discount factors on non-pillar dates required for pricing Swaps are interpolated.

2.4. Tenor Basis Curves

The Tenor Basis Curve is usually calibrated from Tenor Basis Swap. Deposit Rates are occasionally used with Tenor Basis Swap.

Tenor Basis Swap

With the market quote of Tenor Basis Spread, R_{basis}^{tenor} , Tenor Basis Swap is defined with the following equations:

$$\sum_{i_1=0}^{n_1-1} (F^{ref}(T_{i_1}, T_{i_1+1}) + R_{basis}^{tenor}) \tau(T_{i_1}, T_{i_1+1}) P^{OIS}(0, T_{i_1+1}) = \sum_{i_2=0}^{n_2-1} F^{target}(T_{i_2}, T_{i_2+1}) \tau(T_{i_2}, T_{i_2+1}) P^{OIS}(0, T_{i_2+1}).$$

Here the forecast curve to calculate the reference rate and the discount curve are already known, whereas the forecast curve to calculate the target rate is unknown. The solving routine of Tenor Basis Swap is identical to that described in the Swap section above. The discount factors of all maturity tenors are solved simultaneously by using the multivariate Newton Raphson algorithm.

2.5. Cross Currency Basis Curves (USD CSA Curves)

The Cross Currency Basis Curve is calibrated from FX spot, FX Forwards and Cross Currency Basis Swaps, but FX spot and FX Forwards are currently not used in the default setting.

Cross Currency Basis Swap (non-MtM)

To price the Cross Currency Swap, we need to consider the notional exchange at the front and back. So the non Mark-to-Market (non-MtM) Cross Currency Swap is validated as the following equations with the market quote of Cross Currency Spread, R_{basis}^{xccy} :

$$\begin{aligned} -P^{target}(0, T_0) + \sum_{i_1=0}^{n_1-1} (F^{target}(T_{i_1}, T_{i_1+1}) + R_{basis}^{xccy}) \tau(T_{i_1}, T_{i_1+1}) P^{target}(0, T_{i_1+1}) + P^{target}(0, T_{n_1}) \\ = -P_{OIS}^{ref}(0, T_0) + \sum_{i_2=0}^{n_2-1} F^{ref}(T_{i_2}, T_{i_2+1}) \tau(T_{i_2}, T_{i_2+1}) P_{OIS}^{ref}(0, T_{i_2+1}) + P_{OIS}^{ref}(0, T_{n_2}) \end{aligned}$$

Here the forecast curve to calculate the reference rate, the forecast curve to calculate the target rate, and the discount curve to discount the reference leg are already known, and the discount curve to discount the target leg is unknown. When the reference leg is defined in USD, the unknown curve is the discount factors with USD collateral.

Cross Currency Basis Swap (MtM)

Next we move on to price the MtM Cross Currency Swap. The MtM Cross Currency Swap is more popular, because a lot of currencies, predominately G10 currencies, Cross Currency Basis Spreads are quoted based on MtM. The notional adjustments need to be considered on USD leg.

Denoting ccy a non-USD currency, the MtM Cross Currency Swap is validated as the following equations:

$$\begin{aligned}
 & \sum_{i_1=0}^{n_1-1} N^{ccy} (F^{ccy}(T_{i_1}, T_{i_1+1}) + R_{basis}^{xccy}) \tau(T_{i_1}, T_{i_1+1}) P^{ccy_USD\text{CSA}}(0, T_{i_1+1}) \\
 & + N^{ccy} (P^{ccy_USD\text{CSA}}(0, T_{n_1}) - P^{ccy_USD\text{CSA}}(0, T_0)) \\
 & = \sum_{i_2=0}^{n_2-1} N_{i_2}^{USD} F^{USD}(T_{i_2}, T_{i_2+1}) \tau(T_{i_2}, T_{i_2+1}) P_{OIS}^{USD}(0, T_{i_2+1}) \\
 & + \sum_{i_2=0}^{n_2-1} (N_{i_2-1}^{USD} - N_{i_2}^{USD}) P_{OIS}^{USD}(0, T_{i_2+1}) + (N_{i_2-1}^{USD} P_{OIS}^{USD}(0, T_{n_2}) - N_0^{USD} P_{OIS}^{USD}(0, T_0)),
 \end{aligned}$$

where

$$N_{i_2}^{USD} = N^{ccy} \frac{P^{ccy_USD\text{CSA}}(0, T_{i_2})}{P_{OIS}^{USD}(0, T_{i_2})}.$$

The solving routine of Cross Currency Basis Swap, regardless of MtM or non-MtM, is identical to that described in the swap section above. The discount factors of all cross currency basis swap maturity tenors are solved simultaneously by using the multivariate Newton Raphson algorithm.

2.6. FX Forward Constant Curve (Non-USD CSA Curves)

FX Forward Constant Curve is more or less specifically used to create non-USD CSA discount curve. M-Library assumes that FX Forwards are constant and fixed assuming a USD CSA collateral:

$$FwdFX(t, T)^{ccy1/ccy2} = S \frac{P(t, T)^{ccy1_USD\text{CSA}}}{P(t, T)^{ccy2_USD\text{CSA}}}.$$

From which, the discount CSA Curves can be implied as:

$$P(t, T)^{ccy1_ccy2\text{CSA}} = \left(\frac{P(t, T)^{ccy1_USD\text{CSA}}}{P(t, T)^{ccy2_USD\text{CSA}}} \right) \underbrace{\times P(t, T)^{ccy2_ccy2\text{CSA}}}_{P(t, T)_{OIS}^{ccy2}}$$

3. Implementation Test

3.1. Market Data

In this section, we examine the implementation correctness by repricing all the market quotes. We firstly reprice all instruments in M-Library to check the self-consistency, and then independently reprice the same instruments in RMD library to avoid the possibility of a self-consistent mistake in M-Library.

The market data as of 23rd May 2017 is used for the implementation test. The data feed is from Bloomberg, and the source is provided by Quant research & Analytics in Mizuho International plc.

3.2. Self-Consistency in Market Instruments

We firstly reprice all instruments and see if the M-Library yield curves can price them as quoted in the market. Table 3.2.1 to Table 3.2.5 reports the repricing results. From left, it shows the market quotes, par rates estimated from M-Library yield curves, differences between the market quotes and par rates in basis points (bps), and then the PVs of market instruments (with notional of 1,000,000) estimated from M-Library yield curves, PVs' differences from 0 in bps.

USD Self-Consistency

USD OIS					
	Quote	ParRate	+/- Bps	PV	+/- Bps
ON	0.9100%	0.9100%	0.0000	0.0000	0.00000
1W	0.8986%	0.8986%	0.0000	0.0000	0.00000
2W	0.9044%	0.9044%	0.0000	0.0000	0.00000
3W	0.9140%	0.9140%	0.0000	0.0000	0.00000
1M	0.9770%	0.9770%	0.0000	0.0000	0.00000
2M	1.0370%	1.0370%	0.0000	0.0000	0.00000
3M	1.0650%	1.0650%	0.0000	0.0000	0.00000
4M	1.0830%	1.0830%	0.0000	0.0000	0.00000
5M	1.1080%	1.1080%	0.0000	0.0000	0.00000
6M	1.1280%	1.1280%	0.0000	0.0000	0.00000
9M	1.1740%	1.1740%	0.0000	0.0000	0.00000
1Y	1.2120%	1.2120%	0.0000	0.0000	0.00000
2Y	1.3590%	1.3590%	0.0000	-0.0010	-0.00001

* Fixed AN: Float AN

USD Libor-OIS Basis					
	Quote	ParRate	+/- Bps	PV	+/- Bps
3Y	0.1935%	0.1935%	0.0000	0.0000	0.0000
4Y	0.2125%	0.2125%	0.0000	0.0000	0.0000
5Y	0.2256%	0.2256%	0.0000	0.0000	0.0000
7Y	0.2490%	0.2490%	0.0000	0.0001	0.0000
10Y	0.2745%	0.2745%	0.0000	0.0006	0.0000
12Y	0.2885%	0.2885%	0.0000	0.0000	0.0000
15Y	0.3010%	0.3010%	0.0000	0.0000	0.0000
20Y	0.3163%	0.3163%	0.0000	0.0000	0.0000
25Y	0.3298%	0.3298%	0.0000	0.0000	0.0000
30Y	0.3325%	0.3325%	-0.0001	-0.3155	-0.0032

* Float OIS QA: Float Libor QA

USD Swap (*)					
	Quote	ParRate	+/- Bps	PV	+/- Bps
3Y	1.6873%	1.6873%	0.0000	0.0000	0.0000
4Y	1.8012%	1.8012%	0.0000	0.0000	0.0000
5Y	1.8973%	1.8973%	0.0000	0.0000	0.0000
7Y	2.0526%	2.0526%	0.0000	0.0000	0.0000
8Y	2.1155%	2.1155%	0.0000	0.0000	0.0000
9Y	2.1711%	2.1711%	0.0000	0.0000	0.0000
10Y	2.2225%	2.2225%	0.0000	0.0000	0.0000
12Y	2.3043%	2.3043%	0.0000	0.0000	0.0000
15Y	2.3848%	2.3848%	0.0000	0.0000	0.0000
20Y	2.4547%	2.4547%	0.0000	0.0000	0.0000
25Y	2.4814%	2.4814%	0.0000	0.0000	0.0000
30Y	2.4889%	2.4889%	-0.0001	-0.3155	-0.0032

* Fixed SA: Float QA

USD Tenor Basis 3X6					
	Quote	ParRate	+/- Bps	PV	+/- Bps
1Y	0.1530%	0.1530%	0.0000	0.0000	0.0000
2Y	0.1400%	0.1400%	0.0000	0.0000	0.0000
3Y	0.1340%	0.1340%	0.0000	0.0000	0.0000
4Y	0.1330%	0.1330%	0.0000	0.0000	0.0000
5Y	0.1330%	0.1330%	0.0000	0.0000	0.0000
6Y	0.1340%	0.1340%	0.0000	0.0000	0.0000
7Y	0.1360%	0.1360%	0.0000	0.0000	0.0000
8Y	0.1380%	0.1380%	0.0000	0.0000	0.0000
9Y	0.1410%	0.1410%	0.0000	0.0000	0.0000
10Y	0.1430%	0.1430%	0.0000	0.0000	0.0000
12Y	0.1480%	0.1480%	0.0000	0.0000	0.0000
15Y	0.1520%	0.1520%	0.0000	0.0000	0.0000
20Y	0.1550%	0.1550%	0.0000	0.0000	0.0000
25Y	0.1540%	0.1540%	0.0000	0.0000	0.0000
30Y	0.1510%	0.1510%	0.0000	0.0000	0.0000

* Float 3ML QA: Float 6ML SA

USD Future				Convexity Adjustment		
	Quote	Rate	ParRate	+/- Bps	Mean Rev	Volatility
ED1	98.740	1.260%	1.260%	0.0000	3.00%	1.45%
ED2	98.650	1.350%	1.350%	0.0000	3.00%	1.45%
ED3	98.570	1.430%	1.430%	0.0000	3.00%	1.45%
ED4	98.495	1.505%	1.505%	0.0000	3.00%	1.45%
ED5	98.405	1.595%	1.595%	0.0000	3.00%	1.45%
ED6	98.310	1.690%	1.690%	0.0000	3.00%	1.45%
ED7	98.205	1.795%	1.795%	0.0000	3.00%	1.45%
ED8	98.145	1.855%	1.855%	0.0000	3.00%	1.45%

* USD3ML

Table 3.2.1. USD Repricing in M-Library

GBP Self-Consistency

GBP OIS					
	Quote	ParRate	+/- Bps	PV	+/- Bps
ON	0.2112%	0.2130%	0.1800	0.0493	0.0005
1W	0.2130%	0.2130%	0.0000	0.0000	0.0000
1M	0.2130%	0.2130%	0.0000	0.0000	0.0000
2M	0.2120%	0.2120%	0.0000	0.0000	0.0000
3M	0.2150%	0.2150%	0.0000	0.0000	0.0000
4M	0.2180%	0.2180%	0.0000	0.0000	0.0000
5M	0.2190%	0.2190%	0.0000	0.0000	0.0000
6M	0.2230%	0.2230%	0.0000	0.0000	0.0000
7M	0.2280%	0.2280%	0.0000	0.0000	0.0000
8M	0.2310%	0.2310%	0.0000	0.0000	0.0000
9M	0.2350%	0.2350%	0.0000	0.0000	0.0000
10M	0.2400%	0.2400%	0.0000	0.0000	0.0000
11M	0.2430%	0.2430%	0.0000	0.0000	0.0000
1Y	0.2480%	0.2480%	0.0000	0.0000	0.0000
18M	0.2770%	0.2770%	0.0000	0.0000	0.0000
2Y	0.3080%	0.3080%	0.0000	0.0000	0.0000

* Fixed AN: Float AN

GBP Libor-OIS Basis					
	Quote	ParRate	+/- Bps	PV	+/- Bps
3Y	0.1370%	0.1370%	0.0000	0.0001	0.0000
4Y	0.1500%	0.1500%	0.0000	0.0000	0.0000
5Y	0.1610%	0.1610%	0.0000	0.0000	0.0000
7Y	0.1780%	0.1780%	0.0000	0.0000	0.0000
10Y	0.1880%	0.1880%	0.0000	0.0007	0.0000
15Y	0.1820%	0.1820%	0.0000	0.0000	0.0000
20Y	0.1680%	0.1680%	0.0000	0.0000	0.0000
30Y	0.1520%	0.1520%	-0.0001	-0.2535	-0.0025

* Float OIS QA: Float Libor QA

GBP Swap (*)					
	Quote	ParRate	+/- Bps	PV	+/- Bps
2Y	0.4300%	0.4300%	0.0000	0.0000	0.0000
3Y	0.5160%	0.5160%	0.0000	0.0000	0.0000
4Y	0.6030%	0.6030%	0.0000	0.0000	0.0000
5Y	0.6910%	0.6910%	0.0000	0.0000	0.0000
6Y	0.7760%	0.7760%	0.0000	0.0000	0.0000
7Y	0.8600%	0.8600%	0.0000	0.0000	0.0000
8Y	0.9360%	0.9360%	0.0000	0.0000	0.0000
10Y	1.0760%	1.0760%	0.0000	0.0000	0.0000
12Y	1.1880%	1.1880%	0.0000	0.0000	0.0000
15Y	1.3040%	1.3040%	0.0000	0.0000	0.0000
20Y	1.3830%	1.3830%	0.0000	0.0000	0.0000
25Y	1.3890%	1.3890%	0.0000	0.0000	0.0000
30Y	1.3780%	1.3780%	-0.0001	-0.2536	-0.0025

* Fixed SA: Float QA

GBP Tenor Basis 3X6					
	Quote	ParRate	+/- Bps	PV	+/- Bps
1Y	0.1270%	0.1270%	0.0000	0.0000	0.0000
2Y	0.1180%	0.1180%	0.0000	0.0000	0.0000
3Y	0.1130%	0.1130%	0.0000	0.0000	0.0000
4Y	0.1080%	0.1080%	0.0000	0.0000	0.0000
5Y	0.1040%	0.1040%	0.0000	0.0000	0.0000
6Y	0.1000%	0.1000%	0.0000	0.0000	0.0000
7Y	0.0960%	0.0960%	0.0000	0.0000	0.0000
8Y	0.0930%	0.0930%	0.0000	0.0000	0.0000
9Y	0.0900%	0.0900%	0.0000	0.0000	0.0000
10Y	0.0870%	0.0870%	0.0000	-0.0001	0.0000
12Y	0.0810%	0.0810%	0.0000	0.0001	0.0000
15Y	0.0700%	0.0700%	0.0000	0.0002	0.0000
20Y	0.0580%	0.0580%	0.0000	0.0003	0.0000
25Y	0.0500%	0.0500%	0.0000	0.0004	0.0000
30Y	0.0440%	0.0440%	0.0000	0.0007	0.0000

* Float 3ML QA: Float 6ML SA

GBP Cross Currency Basis					
	Quote	ParRate	+/- Bps	PV	+/- Bps
1Y	-0.1450%	-0.1450%	0.0000	0.0000	0.0000
2Y	-0.1375%	-0.1375%	0.0000	0.0000	0.0000
3Y	-0.1300%	-0.1300%	0.0000	0.0000	0.0000
4Y	-0.1225%	-0.1225%	0.0000	0.0000	0.0000
5Y	-0.1075%	-0.1075%	0.0000	0.0000	0.0000
6Y	-0.0950%	-0.0950%	0.0000	0.0000	0.0000
7Y	-0.0850%	-0.0850%	0.0000	0.0000	0.0000
8Y	-0.0750%	-0.0750%	0.0000	0.0000	0.0000
9Y	-0.0700%	-0.0700%	0.0000	0.0000	0.0000
10Y	-0.0650%	-0.0650%	0.0000	0.0000	0.0000
12Y	-0.0675%	-0.0675%	0.0000	0.0000	0.0000
15Y	-0.0750%	-0.0750%	0.0000	0.0000	0.0000
20Y	-0.0850%	-0.0850%	0.0000	0.0000	0.0000
30Y	-0.0400%	-0.0400%	0.0000	0.0000	0.0000

* Float GBP3ML QA: Float USD3ML QA

GBP Future				Convexity Adjustment		
	Quote	Rate	ParRate	+/- Bps	Mean Rev	Volatility
L1	99.685	0.315%	0.315%	0.0000	3.00%	1.20%
L2	99.660	0.340%	0.340%	0.0000	3.00%	1.20%
L3	99.620	0.380%	0.380%	0.0000	3.00%	1.20%
L4	99.580	0.420%	0.420%	0.0000	3.00%	1.20%

*GBP3ML

Table 3.2.2. GBP Repricing in M-Library

EUR Self-Consistency

EUR ECB & OIS					
	Quote	ParRate	+/- Bps	PV	+/- Bps
ECB1	-0.3620%	-0.3620%	0.0000		
ECB2	-0.3620%	-0.3620%	0.0000		
ECB3	-0.3490%	-0.3490%	0.0000		
ECB4	-0.3460%	-0.3460%	0.0000		
ON	-0.3600%	-0.3600%	0.0000	0.0000	0.0000
12M	-0.3360%	-0.3360%	0.0000	0.0000	0.0000
18M	-0.3060%	-0.3060%	0.0000	0.0000	0.0000
2Y	-0.2670%	-0.2670%	0.0000	0.0000	0.0000

* Fixed AN: Float AN

EUR Libor-OIS Basis					
	Quote	ParRate	+/- Bps	PV	+/- Bps
3Y	0.0570%	0.0570%	0.0000	0.6385	0.0064
4Y	0.0660%	0.0660%	0.0000	0.2782	0.0028
5Y	0.0740%	0.0740%	0.0000	-0.6696	-0.0067
6Y	0.0780%	0.0780%	0.0000	-2.2630	-0.0226
7Y	0.0790%	0.0790%	0.0000	-4.4467	-0.0445
8Y	0.0780%	0.0780%	0.0000	-6.9814	-0.0698
9Y	0.0770%	0.0770%	0.0000	-9.8398	-0.0984
10Y	0.0760%	0.0760%	0.0000	-12.9126	-0.1291
11Y	0.0750%	0.0750%	0.0000	-16.0747	-0.1607
12Y	0.0730%	0.0730%	0.0000	-18.9815	-0.1898
15Y	0.0690%	0.0690%	0.0000	-27.2333	-0.2723
20Y	0.0610%	0.0610%	0.0000	-36.2574	-0.3626
25Y	0.0540%	0.0540%	0.0000	-40.8885	-0.4089
30Y	0.0490%	0.0490%	0.0000	-43.9804	-0.4398

* Float OIS QA: Float Libor QA

EUR Swap (*)					
	Quote	ParRate	+/- Bps	PV	+/- Bps
2Y	-0.1320%	-0.1320%	0.0000	0.0000	0.0000
3Y	-0.0210%	-0.0210%	0.0000	0.0000	0.0000
4Y	0.0990%	0.0990%	0.0000	0.0000	0.0000
5Y	0.2250%	0.2250%	0.0000	0.0000	0.0000
6Y	0.3520%	0.3520%	0.0000	0.0000	0.0000
7Y	0.4810%	0.4810%	0.0000	0.0000	0.0000
8Y	0.6080%	0.6080%	0.0000	0.0000	0.0000
9Y	0.7290%	0.7290%	0.0000	0.0000	0.0000
10Y	0.8410%	0.8410%	0.0000	0.0000	0.0000
11Y	0.9420%	0.9420%	0.0000	0.0000	0.0000
12Y	1.0310%	1.0310%	0.0000	0.0000	0.0000
15Y	1.2340%	1.2340%	0.0000	0.0000	0.0000
20Y	1.4090%	1.4090%	0.0000	0.0000	0.0000
25Y	1.4760%	1.4760%	0.0000	0.0000	0.0000
30Y	1.5040%	1.5040%	0.0000	0.0000	0.0000

* Fixed AN: Float SA

EUR Swap (**)					
	Quote	ParRate	+/- Bps	PV	+/- Bps
3Y	-0.1200%	-0.1200%	0.0000	0.0000	0.0000
4Y	-0.0080%	-0.0080%	0.0000	0.0000	0.0000
5Y	0.1110%	0.1110%	0.0000	0.0000	0.0000
6Y	0.2350%	0.2350%	0.0000	0.0000	0.0000
7Y	0.3640%	0.3640%	0.0000	0.0000	0.0000
8Y	0.4930%	0.4930%	0.0000	0.0000	0.0000
9Y	0.6180%	0.6180%	0.0000	0.0000	0.0000
10Y	0.7350%	0.7350%	0.0000	0.0000	0.0000
11Y	0.8410%	0.8410%	0.0000	0.0000	0.0000
12Y	0.9360%	0.9360%	0.0000	0.0000	0.0000
15Y	1.1510%	1.1510%	0.0000	0.0000	0.0000
20Y	1.3440%	1.3440%	0.0000	0.0000	0.0000
25Y	1.4220%	1.4220%	0.0000	0.0000	0.0000
30Y	1.4590%	1.4590%	0.0000	0.0000	0.0000

* Fixed AN: Float QA

EUR Cross Currency Basis					
	Quote	ParRate	+/- Bps	PV	+/- Bps
1Y	-0.3225%	-0.3225%	0.0000	0.0000	0.0000
2Y	-0.3275%	-0.3275%	0.0000	0.0000	0.0000
3Y	-0.3388%	-0.3388%	0.0000	0.0000	0.0000
4Y	-0.3500%	-0.3500%	0.0000	0.0000	0.0000
5Y	-0.3575%	-0.3575%	0.0000	0.0000	0.0000
6Y	-0.3600%	-0.3600%	0.0000	0.0000	0.0000
7Y	-0.3588%	-0.3588%	0.0000	0.0000	0.0000
8Y	-0.3575%	-0.3575%	0.0000	0.0000	0.0000
9Y	-0.3550%	-0.3550%	0.0000	0.0000	0.0000
10Y	-0.3525%	-0.3525%	0.0000	0.0000	0.0000
12Y	-0.3363%	-0.3363%	0.0000	0.0000	0.0000
15Y	-0.2975%	-0.2975%	0.0000	0.0000	0.0000
20Y	-0.2413%	-0.2413%	0.0000	0.0000	0.0000
25Y	-0.1988%	-0.1988%	0.0000	-0.0004	0.0000
30Y	-0.1700%	-0.1700%	0.0000	-0.0007	0.0000

* Float EUR3ML QA: Float USD3ML QA

EUR Future	Convexity Adjustment					
	Quote	Rate	ParRate	+/- Bps	Mean Rev	Volatility
ER1	100.325	-0.325%	-0.325%	0.0000	3.00%	0.90%
ER2	100.310	-0.310%	-0.310%	0.0000	3.00%	0.90%
ER3	100.285	-0.285%	-0.285%	0.0000	3.00%	0.90%
ER4	100.245	-0.245%	-0.245%	0.0000	3.00%	0.90%
ER5	100.200	-0.200%	-0.200%	0.0000	3.00%	0.90%
ER6	100.150	-0.150%	-0.150%	0.0000	3.00%	0.90%
ER7	100.095	-0.095%	-0.095%	0.0000	3.00%	0.90%
ER8	100.045	-0.045%	-0.045%	0.0000	3.00%	0.90%

* EUR3ML

EUR FRA			
	Quote	ParRate	+/- Bps
1X7	-0.247%	-0.247%	0.0000
2X8	-0.238%	-0.238%	0.0000
3X9	-0.229%	-0.229%	0.0000
4X10	-0.219%	-0.219%	0.0000
5X11	-0.207%	-0.207%	0.0000
6X12	-0.194%	-0.194%	0.0000
7X13	-0.181%	-0.181%	0.0000
8X14	-0.166%	-0.166%	0.0000
9X15	-0.149%	-0.149%	0.0000
10X16	-0.134%	-0.134%	0.0000
11X17	-0.116%	-0.116%	0.0000

* EUR6ML

Table 3.2.3. EUR Repricing in M-Library

AUD Self-Consistency

AUD OIS						AUD Swap					
	Quote	ParRate	+/- Bps	PV	+/- Bps		Quote	ParRate	+/- Bps	PV	+/- Bps
ON	1.5000%	1.5000%	0.0000	0.0000	0.0000	1Y (*)	1.7050%	1.7050%	0.0000	0.0000	0.0000
1W	1.5000%	1.5000%	0.0000	0.0000	0.0000	2Y (*)	1.7825%	1.7825%	0.0000	0.0000	0.0000
1M	1.4950%	1.4950%	0.0000	0.0000	0.0000	3Y (*)	1.8875%	1.8875%	0.0000	0.0000	0.0000
2M	1.4913%	1.4913%	0.0000	0.0000	0.0000	1Y (**)	1.8855%	1.8855%	0.0000	0.0000	0.0000
3M	1.4899%	1.4899%	0.0000	0.0000	0.0000	2Y (**)	1.9540%	1.9540%	0.0000	0.0000	0.0000
4M	1.4790%	1.4790%	0.0000	0.0000	0.0000	3Y (**)	2.0580%	2.0580%	0.0000	0.0000	0.0000
5M	1.4930%	1.4930%	0.0000	0.0000	0.0000	4Y (**)	2.1625%	2.1625%	0.0000	0.0000	0.0000
6M	1.4815%	1.4815%	0.0000	0.0000	0.0000	5Y (**)	2.2725%	2.2725%	0.0000	0.0000	0.0000
9M	1.4780%	1.4780%	0.0000	0.0000	0.0000	6Y (**)	2.3788%	2.3787%	0.0000	0.0000	0.0000
1Y	1.4900%	1.4900%	0.0000	-0.0001	0.0000	7Y (**)	2.4788%	2.4787%	0.0000	0.0000	0.0000
2Y	1.5640%	1.5640%	0.0000	0.0000	0.0000	8Y (**)	2.5663%	2.5662%	0.0000	0.0000	0.0000
3Y	1.6660%	1.6660%	0.0000	0.0000	0.0000	9Y (**)	2.6388%	2.6387%	0.0000	0.0000	0.0000
4Y	1.7640%	1.7640%	0.0000	0.0000	0.0000	10Y (**)	2.7050%	2.7050%	0.0000	0.0000	0.0000
5Y	1.8640%	1.8640%	0.0000	0.0000	0.0000	12Y (**)	2.8200%	2.8200%	0.0000	0.0000	0.0000
6Y	1.9640%	1.9640%	0.0000	0.0000	0.0000	15Y (**)	2.9475%	2.9475%	0.0000	0.0000	0.0000
7Y	2.0850%	2.0850%	0.0000	-0.0004	0.0000	20Y (**)	3.0675%	3.0675%	0.0000	0.0000	0.0000
8Y	2.1310%	2.1310%	0.0000	-0.0002	0.0000	25Y (**)	3.1313%	3.1312%	0.0000	0.0000	0.0000
9Y	2.1990%	2.1990%	0.0000	-0.0008	0.0000	30Y (**)	3.1438%	3.1437%	-0.0034	-7.0211	-0.0702
10Y	2.2630%	2.2630%	0.0000	0.0000	0.0000	* Fixed QA: Float QA					
12Y	2.3740%	2.3740%	0.0000	0.0000	0.0000	** Fixed SA: Float SA					
15Y	2.4990%	2.4990%	0.0000	-0.0002	0.0000						
20Y	2.6080%	2.6080%	0.0000	-0.0016	0.0000						
25Y	2.6720%	2.6720%	0.0000	0.0000	0.0000						
30Y	2.6760%	2.6760%	0.0000	0.0000	0.0000						

* Fixed AN: Float AN

AUD Cross Currency Basis						AUD FRA			
	Quote	ParRate	+/- Bps	PV	+/- Bps		Quote	ParRate	+/- Bps
1Y	0.1075%	0.1075%	0.0000	0.0000	0.0000	1x4 (*)	1.713%	1.713%	0.0000
2Y	0.1363%	0.1362%	0.0000	0.0000	0.0000	2x5 (*)	1.708%	1.708%	0.0000
3Y	0.1613%	0.1612%	0.0000	0.0000	0.0000	3x6 (*)	1.699%	1.699%	0.0000
4Y	0.1875%	0.1875%	0.0000	0.0000	0.0000	1x7 (**)	1.883%	1.883%	0.0000
5Y	0.2113%	0.2113%	0.0000	0.0000	0.0000	2x8 (**)	1.878%	1.878%	0.0000
6Y	0.2313%	0.2312%	0.0000	0.0000	0.0000	3x9 (**)	1.878%	1.878%	0.0000
7Y	0.2488%	0.2488%	0.0000	0.0000	0.0000	* AUD3ML			
8Y	0.2625%	0.2625%	0.0000	0.0000	0.0000	** AUD6ML			
9Y	0.2763%	0.2762%	0.0000	0.0000	0.0000				
10Y	0.2875%	0.2875%	0.0000	0.0000	0.0000				
12Y	0.2900%	0.2900%	0.0000	0.0000	0.0000				
15Y	0.2538%	0.2537%	0.0000	0.0000	0.0000				
20Y	0.1513%	0.1512%	0.0000	0.0000	0.0000				
30Y	-0.0525%	-0.0525%	0.0000	0.0000	0.0000				

* Float AUD3ML QA: Float USD3ML QA

Table 3.2.4. AUD Repricing in M-Library

JPY Self-Consistency

JPY OIS					
	Quote	ParRate	+/- Bps	PV	+/- Bps
ON	-0.0510%	-0.0510%	0.0000	0.0000	0.0000
1W	-0.0513%	-0.0512%	0.0000	0.0000	0.0000
1M	-0.0538%	-0.0537%	0.0000	0.0000	0.0000
2M	-0.0550%	-0.0550%	0.0000	0.0000	0.0000
3M	-0.0563%	-0.0562%	0.0000	0.0000	0.0000
4M	-0.0563%	-0.0562%	0.0000	0.0000	0.0000
5M	-0.0563%	-0.0562%	0.0000	0.0000	0.0000
6M	-0.0563%	-0.0562%	0.0000	0.0000	0.0000
9M	-0.0550%	-0.0550%	0.0000	0.0000	0.0000
1Y	-0.0550%	-0.0550%	0.0000	0.0001	0.0000
18M	-0.0513%	-0.0512%	0.0000	0.0001	0.0000
2Y	-0.0488%	-0.0487%	0.0000	0.0001	0.0000

* Fixed AN: Float AN

JPY Libor-OIS Basis					
	Quote	ParRate	+/- Bps	PV	+/- Bps
3Y	0.1000%	0.1000%	0.0000	0.0001	0.0000
4Y	0.1100%	0.1100%	0.0000	-0.1205	-0.0012
5Y	0.1213%	0.1213%	0.0000	-0.3974	-0.0040
6Y	0.1325%	0.1325%	0.0000	-0.8973	-0.0090
7Y	0.1463%	0.1463%	0.0000	-1.7116	-0.0171
8Y	0.1613%	0.1613%	0.0000	-2.9409	-0.0294
9Y	0.1763%	0.1763%	0.0000	-4.6748	-0.0467
10Y	0.1900%	0.1900%	0.0000	-6.9516	-0.0695
12Y	0.2163%	0.2163%	0.0000	-13.7947	-0.1379
15Y	0.2350%	0.2350%	0.0000	-29.0798	-0.2908
20Y	0.2550%	0.2550%	0.0000	-62.8082	-0.6281
25Y	0.2738%	0.2738%	0.0000	-98.1878	-0.9819
30Y	0.2900%	0.2900%	0.0000	-132.9513	-1.3295

* Float OIS QA: Float Libor SA

JPY Swap					
	Quote	ParRate	+/- Bps	PV	+/- Bps
1Y	0.0313%	0.0312%	0.0000	0.0000	0.0000
18M	0.0363%	0.0362%	0.0000	0.0000	0.0000
2Y	0.0413%	0.0412%	0.0000	0.0000	0.0000
3Y	0.0500%	0.0500%	0.0000	0.0000	0.0000
4Y	0.0663%	0.0662%	0.0000	0.0000	0.0000
5Y	0.0875%	0.0875%	0.0000	0.0000	0.0000
6Y	0.1125%	0.1125%	0.0000	0.0000	0.0000
7Y	0.1413%	0.1412%	0.0000	0.0000	0.0000
8Y	0.1738%	0.1737%	0.0000	0.0000	0.0000
9Y	0.2088%	0.2088%	0.0000	0.0000	0.0000
10Y	0.2450%	0.2450%	0.0000	0.0000	0.0000
12Y	0.3263%	0.3263%	0.0000	0.0000	0.0000
15Y	0.4575%	0.4575%	0.0000	0.0000	0.0000
20Y	0.6413%	0.6413%	0.0000	0.0000	0.0000
25Y	0.7450%	0.7450%	0.0000	0.0000	0.0000
30Y	0.8038%	0.8038%	0.0000	0.0000	0.0000

* Fixed SA: Float SA

JPY Tenor Basis 3X6					
	Quote	ParRate	+/- Bps	PV	+/- Bps
1Y	0.0250%	0.0250%	0.0000	0.0000	0.0000
18M	0.0250%	0.0250%	0.0000	0.0000	0.0000
2Y	0.0263%	0.0262%	0.0000	0.0000	0.0000
3Y	0.0288%	0.0287%	0.0000	0.0000	0.0000
4Y	0.0313%	0.0312%	0.0000	0.0000	0.0000
5Y	0.0350%	0.0350%	0.0000	0.0000	0.0000
6Y	0.0400%	0.0400%	0.0000	0.0000	0.0000
7Y	0.0463%	0.0462%	0.0000	0.0000	0.0000
8Y	0.0550%	0.0550%	0.0000	0.0000	0.0000
9Y	0.0638%	0.0638%	0.0000	0.0001	0.0000
10Y	0.0713%	0.0713%	0.0000	0.0001	0.0000
12Y	0.0825%	0.0825%	0.0000	0.0001	0.0000
15Y	0.0900%	0.0900%	0.0000	0.0000	0.0000
20Y	0.0975%	0.0975%	0.0000	-0.0001	0.0000
25Y	0.1113%	0.1112%	0.0000	-0.0002	0.0000
30Y	0.1225%	0.1225%	0.0000	-0.0002	0.0000

* Float 3ML QA: Float 6ML SA

JPY Cross Currency Basis					
	Quote	ParRate	+/- Bps	PV	+/- Bps
1Y	-0.4800%	-0.4800%	0.0009	0.0898	0.0009
18M	-0.5138%	-0.5137%	0.0009	0.1341	0.0013
2Y	-0.5650%	-0.5650%	0.0008	0.1605	0.0016
3Y	-0.6313%	-0.6312%	0.0005	0.1661	0.0017
4Y	-0.6800%	-0.6800%	0.0004	0.1662	0.0017
5Y	-0.7150%	-0.7150%	0.0003	0.1662	0.0017
6Y	-0.7363%	-0.7362%	0.0003	0.1662	0.0017
7Y	-0.7475%	-0.7475%	0.0002	0.1662	0.0017
8Y	-0.7488%	-0.7487%	0.0002	0.1662	0.0017
9Y	-0.7413%	-0.7412%	0.0002	0.1662	0.0017
10Y	-0.7288%	-0.7287%	0.0002	0.1662	0.0017
12Y	-0.6963%	-0.6962%	0.0001	0.1662	0.0017
15Y	-0.6463%	-0.6462%	0.0001	0.1662	0.0017
20Y	-0.5800%	-0.5800%	0.0001	0.1662	0.0017
25Y	-0.5313%	-0.5312%	0.0001	0.1662	0.0017
30Y	-0.4888%	-0.4888%	-0.0001	-0.4484	-0.0045

* Float JPY3ML QA: Float USD3ML QA

JPY FRA			
	Quote	ParRate	+/- Bps
1x7 (*)	0.028%	0.028%	0.0000
2x8 (*)	0.031%	0.031%	0.0000
3x9 (*)	0.033%	0.033%	0.0000
1x4 (**)	-0.006%	-0.006%	0.0000
2x5 (**)	0.000%	0.000%	0.0000
3x6 (**)	0.006%	0.006%	0.0000

* JPY6ML

** JPY3ML

Table 3.2.5. JPY Repricing in M-Library

This test has been passed. The par rates are sufficiently close to all the market quotes, with accuracy greater than or equal to 1/10,000th of a basis point for almost all of the instruments. The only instrument to be noted is that AUD 30Y Swap is the order of 1/1000th basis point, but we suppose it is not a problem, as it is in the bid-offer spread.

3.3. Independent repricing of Market Instruments

The test above is sufficient to prove the validity of the curve construction. Nevertheless, in order to avoid the possibility of a self-consistent mistake in M-Library, we secondly reprice the exact same instruments in RMD library.

In this test, all M-Library yield curves are brought into RMD library as monthly zero rates. Then OIS curve is spline interpolated on zero rates, whereas Swap and Basis Curves are spline interpolated on logarithm of discount factors. Due to the restriction of RMD library, this setting is slightly different from M-Library's setting in Table 2.1. Table 3.3.1 to Table 3.2.5 reports the repricing results. The left column shows the PVs of market instruments (with notional of 1,000,000) estimated from M-Library yield curves, and the right column shows the PVs' differences from 0 in basis points (bps).

USD Independent Repricing

USD OIS			USD Libor-OIS Basis			USD Swap (*)			USD Tenor Basis 3X6		
	PV	+/- Bps		PV	+/- Bps		PV	+/- Bps		PV	+/- Bps
1W	10.3531	0.1035	3Y	2.1445	0.0214	3Y	-1.1249	-0.0112	1Y	1.2118	0.0121
2W	18.4478	0.1845	4Y	2.6328	0.0263	4Y	-1.1219	-0.0112	2Y	0.5238	0.0052
3W	22.0700	0.2207	5Y	3.1909	0.0319	5Y	-1.1206	-0.0112	3Y	1.1808	0.0118
1M	-20.8312	-0.2083	7Y	4.4420	0.0444	7Y	-1.1169	-0.0112	4Y	1.1437	0.0114
2M	7.3883	0.0739	10Y	6.3898	0.0639	8Y	-1.1543	-0.0115	5Y	1.1438	0.0114
3M	-8.9343	-0.0893	12Y	7.7387	0.0774	9Y	-1.1515	-0.0115	6Y	1.1438	0.0114
4M	2.4886	0.0249	15Y	9.6507	0.0965	10Y	-1.1518	-0.0115	7Y	1.1430	0.0114
5M	-5.9881	-0.0599	20Y	12.4440	0.1244	12Y	-1.1511	-0.0115	8Y	1.0992	0.0110
6M	0.6141	0.0061	25Y	14.6229	0.1462	15Y	-1.1336	-0.0113	9Y	1.0980	0.0110
9M	-3.1133	-0.0311	30Y	17.8567	0.1786	20Y	-1.1163	-0.0112	10Y	1.0987	0.0110
1Y	-2.5015	-0.0250	* Float OIS QA: Float Libor QA			25Y	-1.1419	-0.0114	12Y	1.0977	0.0110
2Y	-2.8691	-0.0287				30Y	-1.7107	-0.0171	15Y	1.1366	0.0114
* Fixed AN: Float AN						* Fixed SA: Float QA			20Y	1.0722	0.0107
									25Y	1.0983	0.0110
									30Y	2.2393	0.0224
									* Float 3ML QA: Float 6ML SA		

Table 3.3.1. USD Repricing in RMD Library

GBP Independent Repricing

GBP OIS			GBP Libor-OIS Basis			GBP Swap (*)			GBP Tenor Basis 3X6		
	PV	+/- Bps		PV	+/- Bps		PV	+/- Bps		PV	+/- Bps
1W	-0.0001	0.0000	3Y	0.0630	0.0006	2Y	0.0000	0.0000	1Y	0.0000	0.0000
1M	-0.0005	0.0000	4Y	0.1200	0.0012	3Y	-0.0004	0.0000	2Y	0.0000	0.0000
2M	0.1040	0.0010	5Y	0.2159	0.0022	4Y	-0.0061	-0.0001	3Y	0.0001	0.0000
3M	-0.0015	0.0000	7Y	0.5207	0.0052	5Y	-0.0061	-0.0001	4Y	-0.0159	-0.0002
4M	-0.0641	-0.0006	10Y	1.2624	0.0126	6Y	-0.0061	-0.0001	5Y	-0.0159	-0.0002
5M	-0.0026	0.0000	15Y	2.8673	0.0287	7Y	-0.0061	-0.0001	6Y	-0.0159	-0.0002
6M	-0.0031	0.0000	20Y	4.1513	0.0415	8Y	-0.0059	-0.0001	7Y	-0.0159	-0.0002
7M	-0.1237	-0.0012	30Y	14.4788	0.1448	10Y	-0.0059	-0.0001	8Y	-0.0168	-0.0002
8M	-0.0049	0.0000	* Float OIS QA: Float Libor QA			12Y	-0.0061	-0.0001	9Y	-0.0169	-0.0002
9M	-0.0056	-0.0001				15Y	0.0169	0.0002	10Y	-0.0169	-0.0002
10M	-0.0063	-0.0001				20Y	-0.0014	0.0000	12Y	-0.0047	0.0000
11M	-0.0076	-0.0001				25Y	-0.0013	0.0000	15Y	-0.0275	-0.0003
1Y	-0.0086	-0.0001				30Y	8.7755	0.0878	20Y	-0.0198	-0.0002
18M	-0.0156	-0.0002	* Fixed SA: Float QA						25Y	-0.0197	-0.0002
2Y	-0.0269	-0.0003							30Y	-9.0281	-0.0903
* Fixed AN: Float AN									* Float 3ML QA: Float 6ML SA		

GBP Cross Currency Basis		
	PV	+/- Bps
1Y	-4.2725	-0.0427
2Y	-4.1787	-0.0418
3Y	-2.8735	-0.0287
4Y	-2.2443	-0.0224
5Y	-3.4236	-0.0342
6Y	-6.7492	-0.0675
7Y	-8.5487	-0.0855
8Y	-7.0105	-0.0701
9Y	-4.6393	-0.0464
10Y	-4.3787	-0.0438
12Y	-5.6253	-0.0563
15Y	-5.6249	-0.0562
20Y	-5.5090	-0.0551
30Y	-15.4375	-0.1544

* Float GBP3ML QA: Float USD3ML QA

Table 3.3.2. GBP Repricing in RMD Library

EUR Independent Repricing

EUR OIS	PV	+/- Bps
12M	0.1038	0.0010
18M	0.1048	0.0010
2Y	0.1038	0.0010

* Fixed AN: Float AN

EUR Libor-OIS Basis	PV	+/- Bps
3Y	-0.7626	-0.0076
4Y	-0.7621	-0.0076
5Y	-0.7630	-0.0076
6Y	-0.7622	-0.0076
7Y	-0.7626	-0.0076
8Y	-0.7686	-0.0077
9Y	-0.7701	-0.0077
10Y	-0.7685	-0.0077
11Y	-0.7701	-0.0077
12Y	-0.7693	-0.0077
15Y	-0.7703	-0.0077
20Y	-0.7579	-0.0076
25Y	-0.7694	-0.0077
30Y	3.9825	0.0398

* Float OIS QA: Float Libor QA

EUR Swap (*)	PV	+/- Bps
2Y	0.2707	0.0027
3Y	0.2405	0.0024
4Y	0.2410	0.0024
5Y	0.2407	0.0024
6Y	0.2408	0.0024
7Y	0.2418	0.0024
8Y	0.2154	0.0022
9Y	0.2157	0.0022
10Y	0.2156	0.0022
11Y	0.2158	0.0022
12Y	0.2147	0.0021
15Y	0.2296	0.0023
20Y	0.2174	0.0022
25Y	0.2174	0.0022
30Y	-6.3805	-0.0638

* Fixed AN: Float SA

EUR Swap (**)	PV	+/- Bps
3Y	-0.6580	-0.0066
4Y	-0.6573	-0.0066
5Y	-0.6575	-0.0066
6Y	-0.6573	-0.0066
7Y	-0.6566	-0.0066
8Y	-0.6686	-0.0067
9Y	-0.6684	-0.0067
10Y	-0.6688	-0.0067
11Y	-0.6682	-0.0067
12Y	-0.6691	-0.0067
15Y	-0.6691	-0.0067
20Y	-0.6562	-0.0066
25Y	-0.6665	-0.0067
30Y	-10.9520	-0.1095

** Fixed AN: Float QA

EUR Cross Currency Basis	PV	+/- Bps
1Y	4.2496	0.0425
2Y	11.2334	0.1123
3Y	15.4571	0.1546
4Y	16.9772	0.1698
5Y	21.7376	0.2174
6Y	26.7963	0.2680
7Y	31.5375	0.3154
8Y	36.1631	0.3616
9Y	39.9308	0.3993
10Y	41.1398	0.4114
12Y	44.2842	0.4428
15Y	44.8363	0.4484
20Y	42.3109	0.4231
25Y	40.6776	0.4068
30Y	20.0751	0.2008

* Float EUR3ML QA: Float USD3ML QA

Table 3.3.3. EUR Repricing in RMD Library

AUD Independent Repricing

AUD OIS	PV	+/- Bps
1W	-0.9209	-0.0092
1M	0.7799	0.0078
2M	-0.0694	-0.0007
3M	0.3547	0.0035
4M	-0.1717	-0.0017
5M	0.3123	0.0031
6M	0.0330	0.0003
9M	0.1317	0.0013
1Y	0.1367	0.0014
2Y	0.1367	0.0014
3Y	0.1355	0.0014
4Y	0.1368	0.0014
5Y	0.1360	0.0014
6Y	0.1382	0.0014
7Y	0.1340	0.0013
8Y	0.1613	0.0016
9Y	0.1311	0.0013
10Y	0.1371	0.0014
12Y	0.1369	0.0014
15Y	0.1369	0.0014
20Y	0.1355	0.0014
25Y	0.1355	0.0014
30Y	2.8235	0.0282

* Fixed AN: Float AN

AUD Swap	PV	+/- Bps
1Y (*)	-3.7977	-0.0380
2Y (*)	-3.7977	-0.0380
3Y (*)	-3.7655	-0.0377
1Y (**)	-7.1066	-0.0711
2Y (**)	-7.1068	-0.0711
3Y (**)	-7.1067	-0.0711
4Y (**)	-7.1066	-0.0711
5Y (**)	-7.1093	-0.0711
6Y (**)	-7.1064	-0.0711
7Y (**)	-7.1070	-0.0711
8Y (**)	-7.1071	-0.0711
9Y (**)	-7.0465	-0.0705
10Y (**)	-7.0978	-0.0710
12Y (**)	-7.0993	-0.0710
15Y (**)	-4.6719	-0.0467
20Y (**)	-7.0790	-0.0708
25Y (**)	-7.0690	-0.0707
30Y (**)	-7.0699	-0.0707

* Fixed QA: Float QA

** Fixed SA: Float SA

AUD Cross Currency Basis	PV	+/- Bps
1Y	7.2500	0.0725
2Y	6.8810	0.0688
3Y	9.9094	0.0991
4Y	9.9075	0.0991
5Y	8.0700	0.0807
6Y	3.3551	0.0336
7Y	3.1320	0.0313
8Y	5.1283	0.0513
9Y	5.7548	0.0575
10Y	5.7400	0.0574
12Y	4.9574	0.0496
15Y	5.7566	0.0576
20Y	3.8415	0.0384
30Y	34.8210	0.3482

* Float AUD3ML QA: Float USD3ML QA

Table 3.3.4. AUD Repricing in RMD Library

JPY Independent Repricing

JPY OIS	PV	+/- Bps
ON		0.0000
1W	-0.4101	-0.0041
1M	0.2972	0.0030
2M	0.0830	0.0008
3M	0.1435	0.0014
4M	0.1291	0.0013
5M	0.1306	0.0013
6M	0.1369	0.0014
9M	0.1278	0.0013
1Y	0.1317	0.0013
18M	0.1300	0.0013
2Y	0.1311	0.0013

* Fixed AN: Float AN

JPY Libor-OIS Basis	PV	+/- Bps
3Y	0.5059	0.0051
4Y	0.5071	0.0051
5Y	0.5061	0.0051
6Y	0.5071	0.0051
7Y	0.5022	0.0050
8Y	0.4954	0.0050
9Y	0.4944	0.0049
10Y	0.4960	0.0050
12Y	0.4943	0.0049
15Y	0.5148	0.0051
20Y	0.4988	0.0050
25Y	0.4962	0.0050
30Y	-10.7192	-0.1072

* Float OIS QA: Float Libor SA

JPY Swap	PV	+/- Bps
1Y	0.6386	0.0064
18M	0.6414	0.0064
2Y	0.6382	0.0064
3Y	0.6393	0.0064
4Y	0.6377	0.0064
5Y	0.6378	0.0064
6Y	0.6377	0.0064
7Y	0.6383	0.0064
8Y	0.6275	0.0063
9Y	0.6270	0.0063
10Y	0.6275	0.0063
12Y	0.6268	0.0063
15Y	0.6444	0.0064
20Y	0.6249	0.0062
25Y	0.6250	0.0063
30Y	-36.8836	-0.3688

* Fixed SA: Float SA

JPY Tenor Basis 3X6	PV	+/- Bps
1Y	0.6724	0.0067
18M	0.6745	0.0067
2Y	0.6734	0.0067
3Y	0.6716	0.0067
4Y	0.6706	0.0067
5Y	0.6706	0.0067
6Y	0.6704	0.0067
7Y	0.6715	0.0067
8Y	0.6646	0.0066
9Y	0.6647	0.0066
10Y	0.6649	0.0066
12Y	0.6641	0.0066
15Y	0.6743	0.0067
20Y	0.6454	0.0065
25Y	0.6600	0.0066
30Y	-3.5456	-0.0355

* Float 3ML QA: Float 6ML SA

JPY Cross Currency Basis		
	PV	+/- Bps
1Y	0.4210	0.0042
18M	-0.0478	-0.0005
2Y	1.5624	0.0156
3Y	0.9396	0.0094
4Y	0.9435	0.0094
5Y	1.7797	0.0178
6Y	2.8128	0.0281
7Y	2.8195	0.0282
8Y	2.7730	0.0277
9Y	2.8496	0.0285
10Y	2.8507	0.0285
12Y	4.9293	0.0493
15Y	8.4064	0.0841
20Y	10.1966	0.1020
25Y	8.9998	0.0900
30Y	-14.3940	-0.1439

* Float JPY3ML QA: Float USD3ML QA

Table 3.3.5. JPY Repricing in RMD Library

PVs of some instruments were slightly away from par value (i.e. 0). We found that the discrepancy in repricing was observed when M-Library and our library used different cashflow schedule. M-Library does not hold fixing date information. For interest rate, M-Library uses the accrual start date instead of fixing date + spot lag, so the actual differences between the accrual start date and fixing date + spot lag may be ignored, if any. And for FX, M-Library uses the payment date instead of fixing date + spot lag, so again, the actual differences between the payment date and fixing date + spot lag may be ignored, if any. The differences were occasionally caused by holiday adjustments or differences between trades' specific fixing calendars and market common conventions of the instruments.

However, the PVs were generally under 0.2bp for 30Y, and even the worst case, AUD Cross Currency Basis and JPY Swap, have the PVs under 0.4bp for 30Y. Therefore, we evaluated this test had been passed as well.

4. Modelling Test

4.1. Smoothness of Forward Curves

The M-Library forward curves are shown below. Left top charts are the forward curves up to 30 years, and right top charts are the forward curves up to 10 years.

As we find the forward OIS curves tend to have:

- ✓ Zigzag shape at short end,
- ✓ Curtain-like shape at long end,

left middle charts represent the forward OIS curves only up to 30 years, and right middle charts represent the forward OIS curves up to 3 years only. In right middle charts, grey lines indicate the pillar dates, that are equal to maturity tenors of given OIS outrights.

Left bottom charts represent the forward 3M Libor curves up to 3 years, in which grey lines indicate the future contract roles to see the stableness of the Swap Curves.

USD Forward Curves

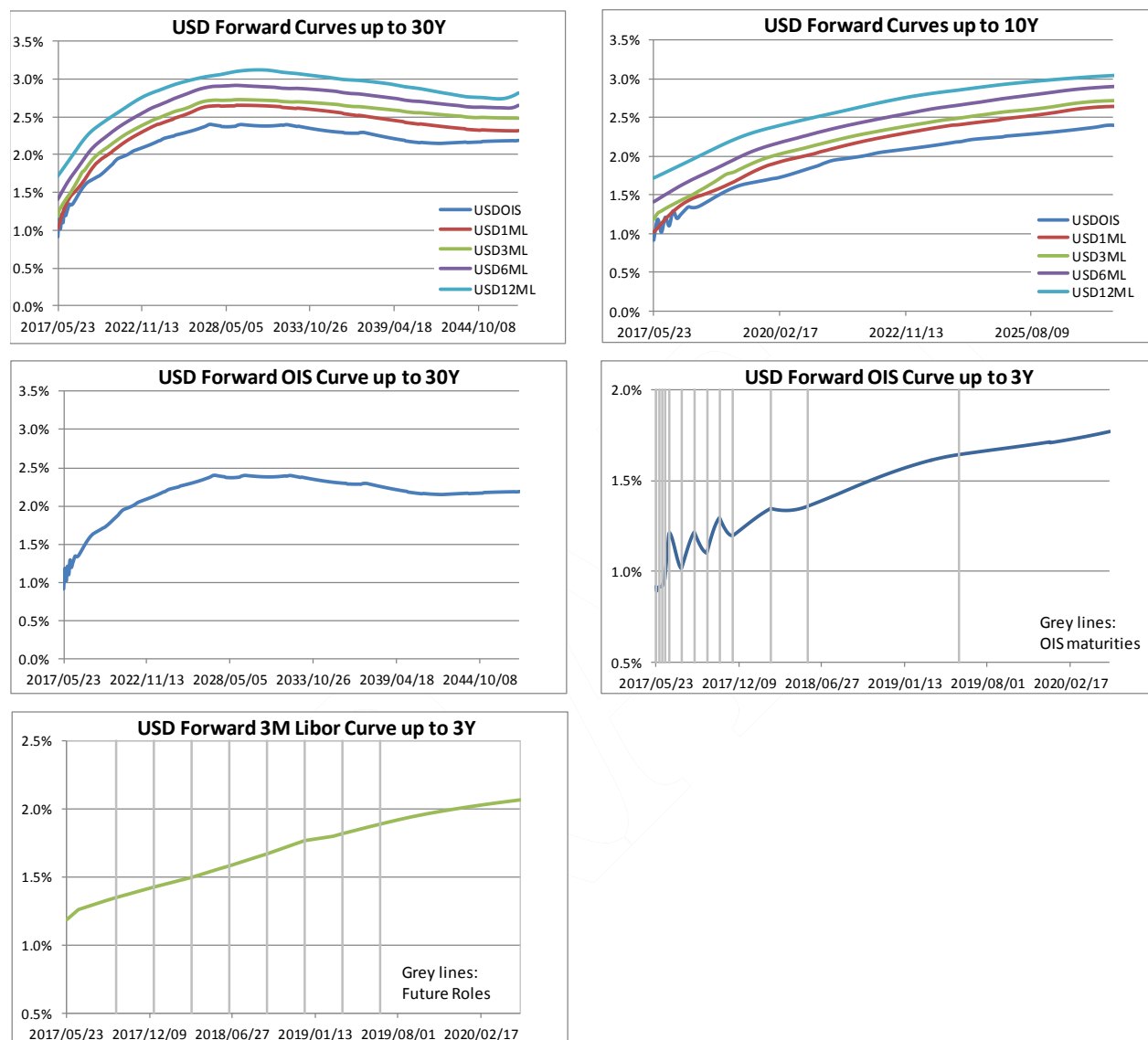


Figure 4.1.1 USD Forward Curves

GBP Forward Curves

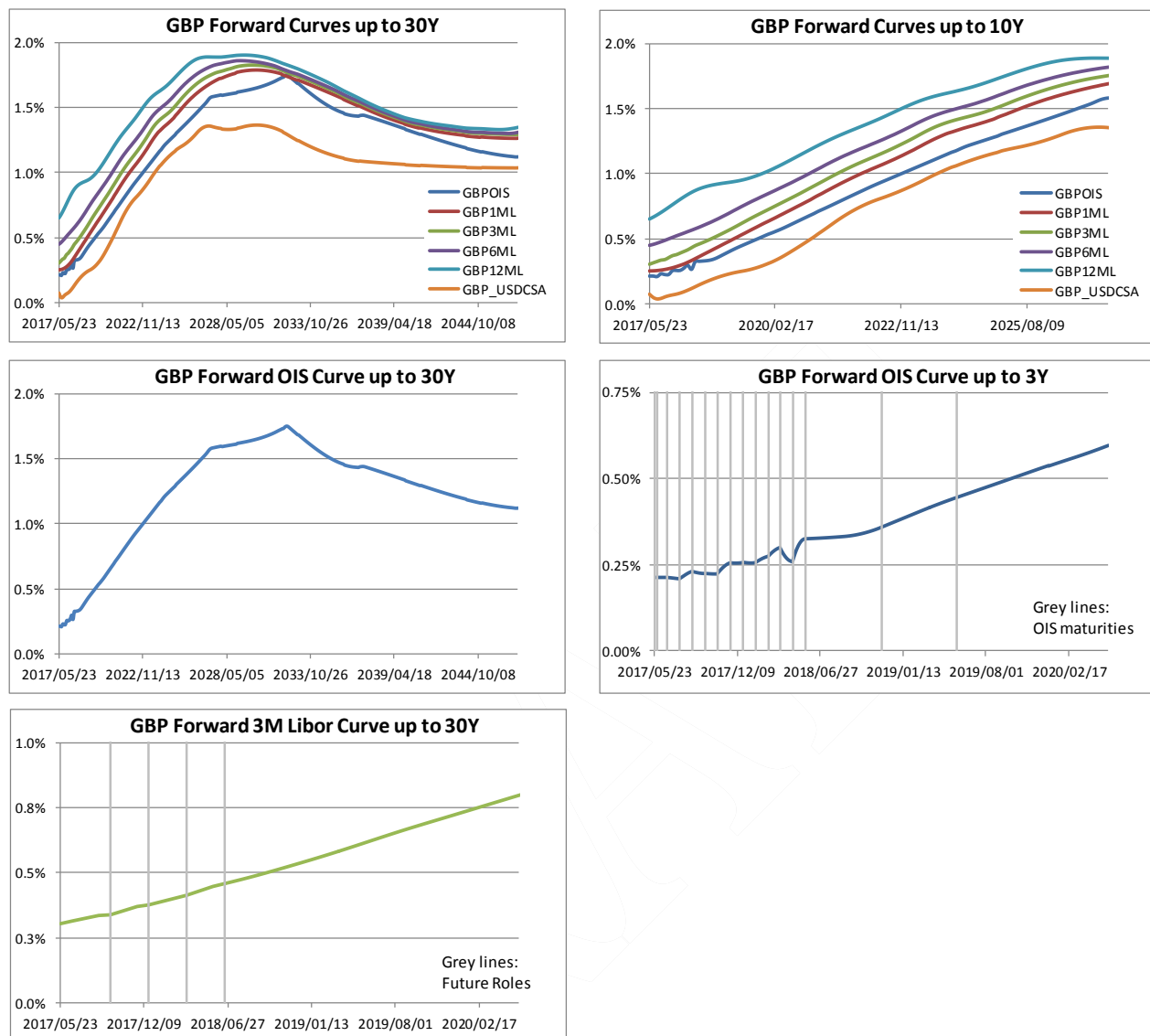
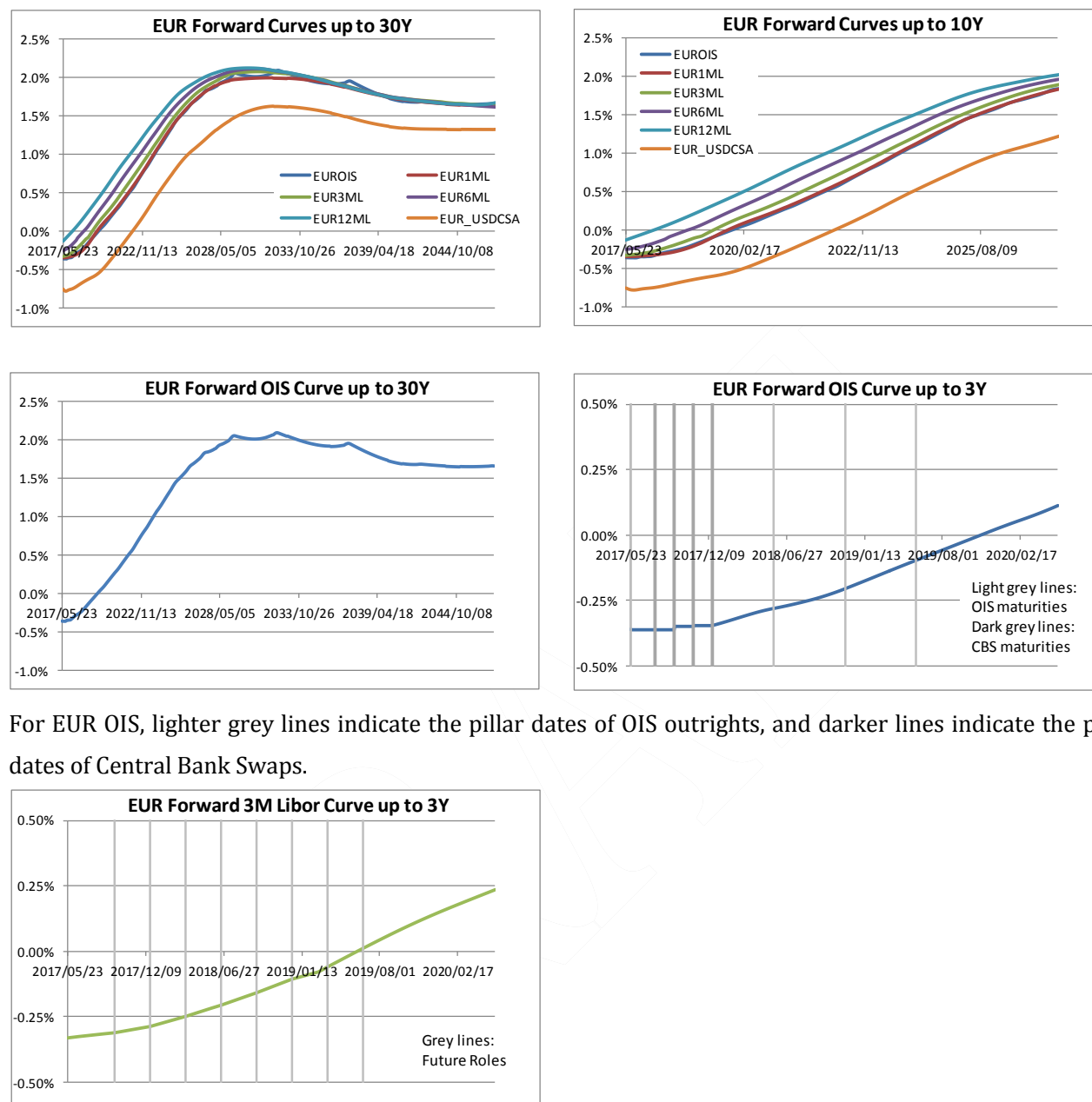


Figure 4.1.2 GBP Forward Curves

EUR Forward Curves



For EUR OIS, lighter grey lines indicate the pillar dates of OIS outright, and darker lines indicate the pillar dates of Central Bank Swaps.

Figure 4.1.3 EUR Forward Curves

AUD Forward Curves

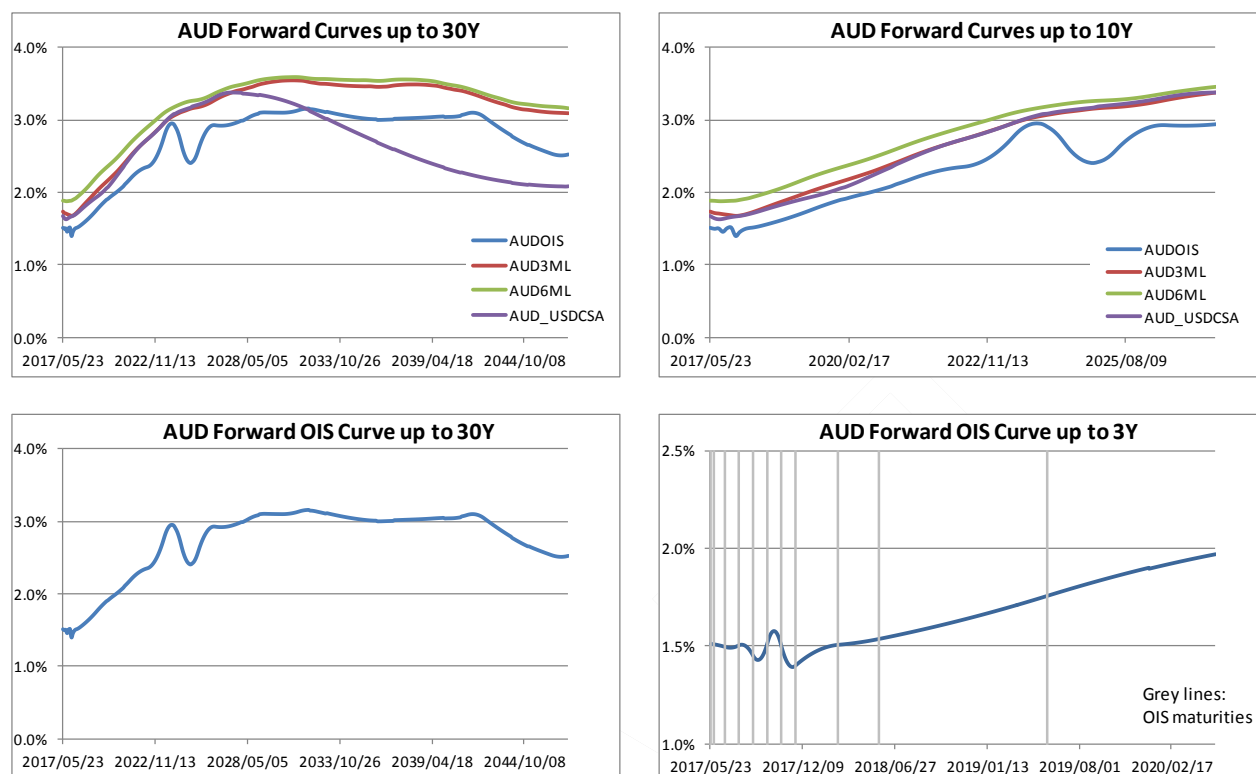


Figure 4.1.4 AUD Forward Curves

JPY Forward Curves

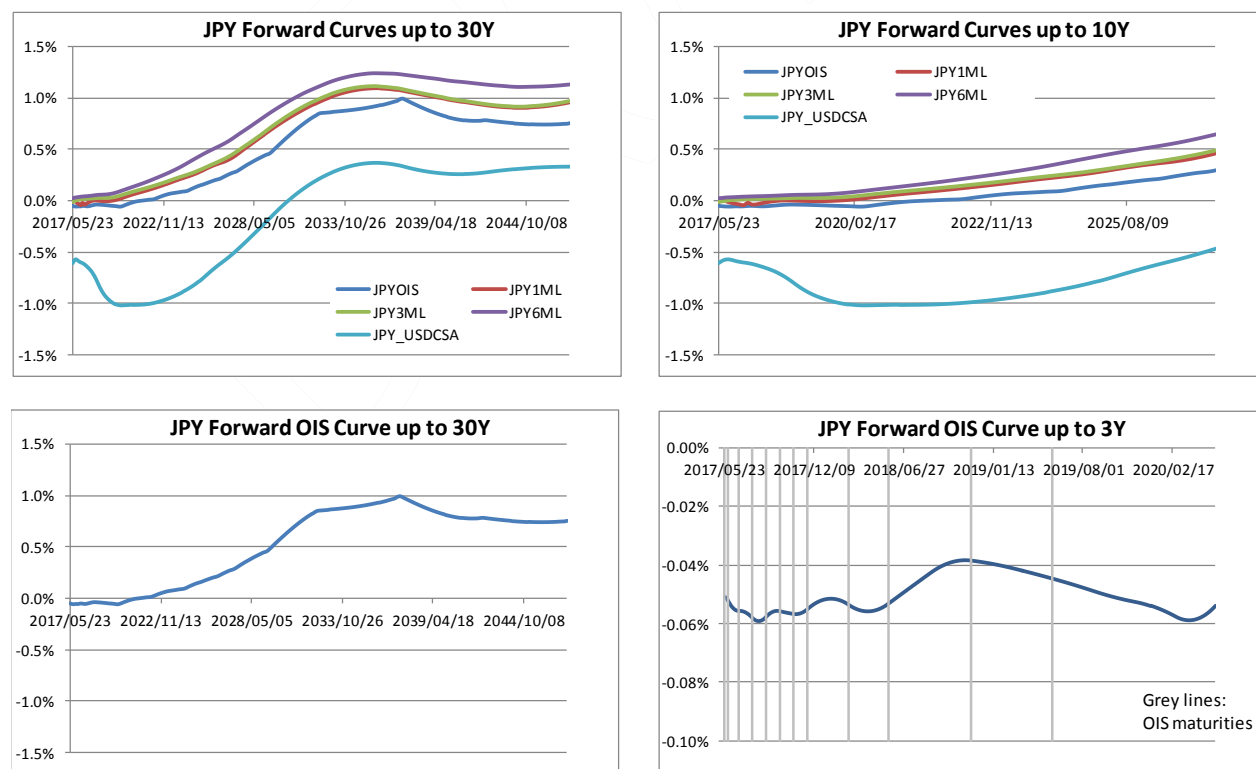


Figure 4.1.5 JPY Forward Curves

The forward curves are generally smooth, except for OIS Curves. As you can see in the forward 3M Libor curves, it is especially worth mentioning that the Swap Curves remain stable during futures contract roles. As described in the beginning of 4.1, the forward OIS curves tend to have zigzag shape at short end and curtain-like shape at long end. However, by considering pillar dates, it seems natural to have zigzag shape at short end. The short end should become smoother if a certain OIS outrights are selected to build curves.

For example, AUD OIS curve has become smoother by excluding 7Y OIS.

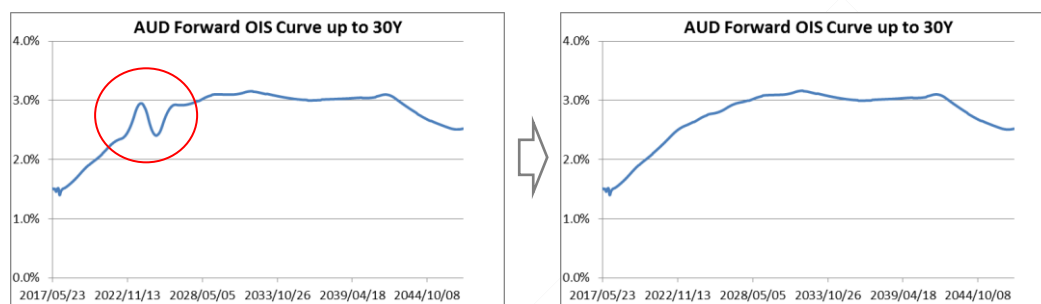


Figure 4.1.6 AUD Forward OIS Curve

For the curtain-like shape, it will be discussed later in 4.3. In either case, we think these shapes in OIS Curves are not critical in terms of risk management, because we confirmed Overnight Index Swap were repriced correctly, and other than Overnight Index Swap, the OIS Curve is used for discounting only.

4.2. Simultaneous Optimization and Iterative Bootstrap

As described in Chapter 2, M-Library uses Newton-Raphson algorithm, in which the discount factors are found sequentially for a curve. As an alternative approach, the iterative bootstrap is used in RMD library.

To describe the iterative bootstrap, let $P^{(k)}(0, t_i)$ is the value for $P(0, t_i)$ found in the k -th iteration. Then the algorithm is:

1. Make initial guesses for $P^{(0)}(0, t_j)$, $j = 1, \dots, N$
2. Set the k -th iteration as below:
 - a. $P^{(k)}(0, t_j)$, $j = 1, \dots, i - 1$ and $P^{(k-1)}(0, t_i)$ are all known
 - b. Compute $P^{(k)}(0, t_i) = \text{solve} \left(F \left(P^{(k)}(0, t_1), \dots, P^{(k)}(0, t_{i-1}), P^{(k-1)}(0, t_i) \right) \right)$
 - c. Repeat above for $i = 1, \dots, N$
3. If $\max_i |P^{(k)}(0, t_i) - P^{(k-1)}(0, t_i)|$ is less than a criteria, stop. Otherwise, return to Step 2.

4.3. JPY OIS, Swap, Tenor Basis and Cross Currency Basis Curves from Iterative Bootstrap

The above iterative bootstrap is applied to JPY curves on 23rd May 2017. The same market instruments as 3.1. are used to construct JPY OIS, 6M Libor, 3M Libor Curves and JPY with USD CSA Collateral Curves. The forward curves are shown below together with the discount factor and zero rate curves.

JPY OIS Curves from Iterative Bootstrap

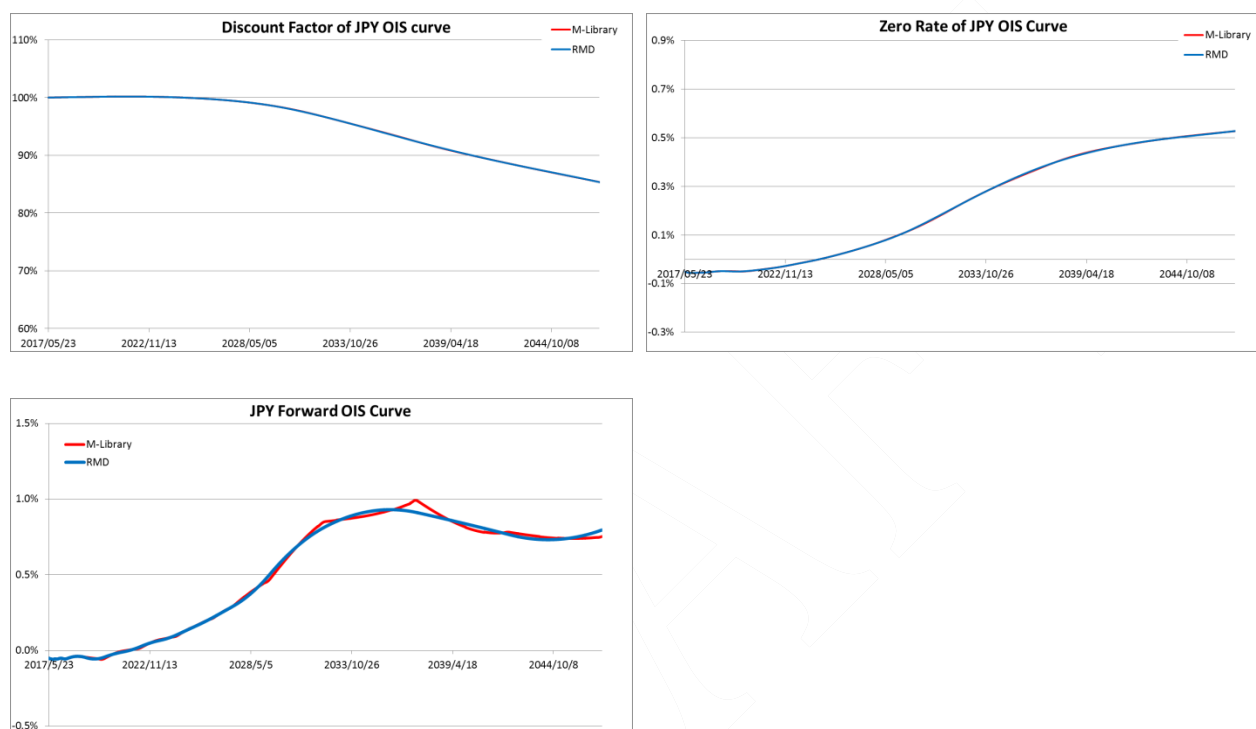


Figure 4.3.1 JPY OIS Curves

JPY 6M Libor Curves from Iterative Bootstrap

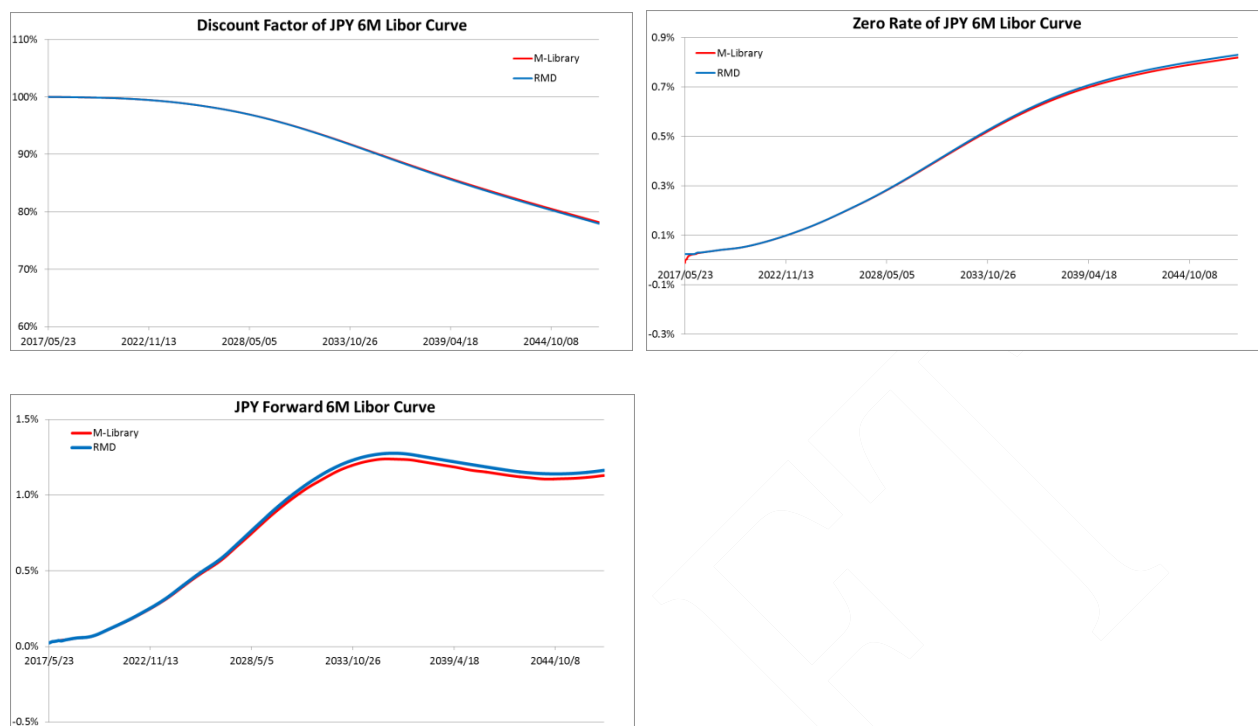


Figure 4.3.2 JPY 6M Libor Curves

JPY 3M Libor Curves from Interactive Bootstrap

JPY 3M Libor Curve is derived from Tenor Basis Swaps.

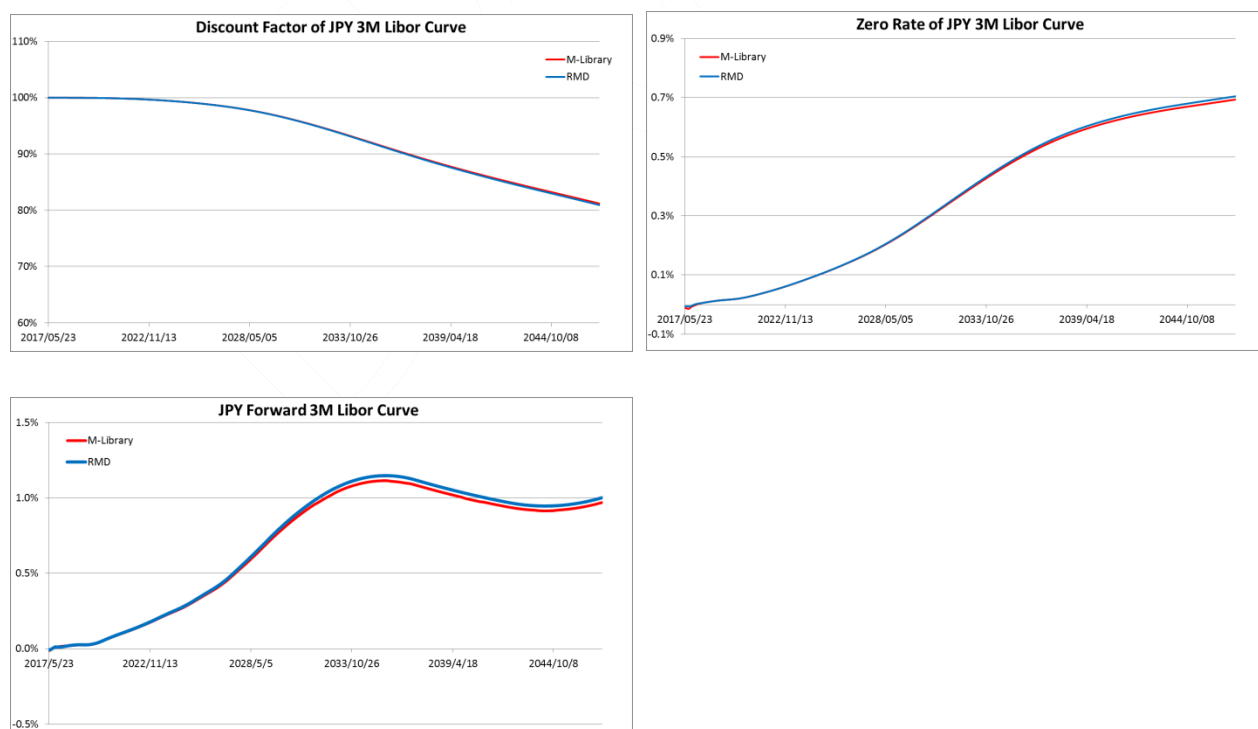


Figure 4.3.3 JPY 3M Libor Curves

JPY with USD CSA Collateral Curves from Interactive Bootstrap

JPY with USD CSA collateral curves are derived from JPY Libor 3M (forecast curve to the target rate), USD Libor 3M (forecast curve to the reference rate) and USD OIS (discount curve to the reference leg) Curves.

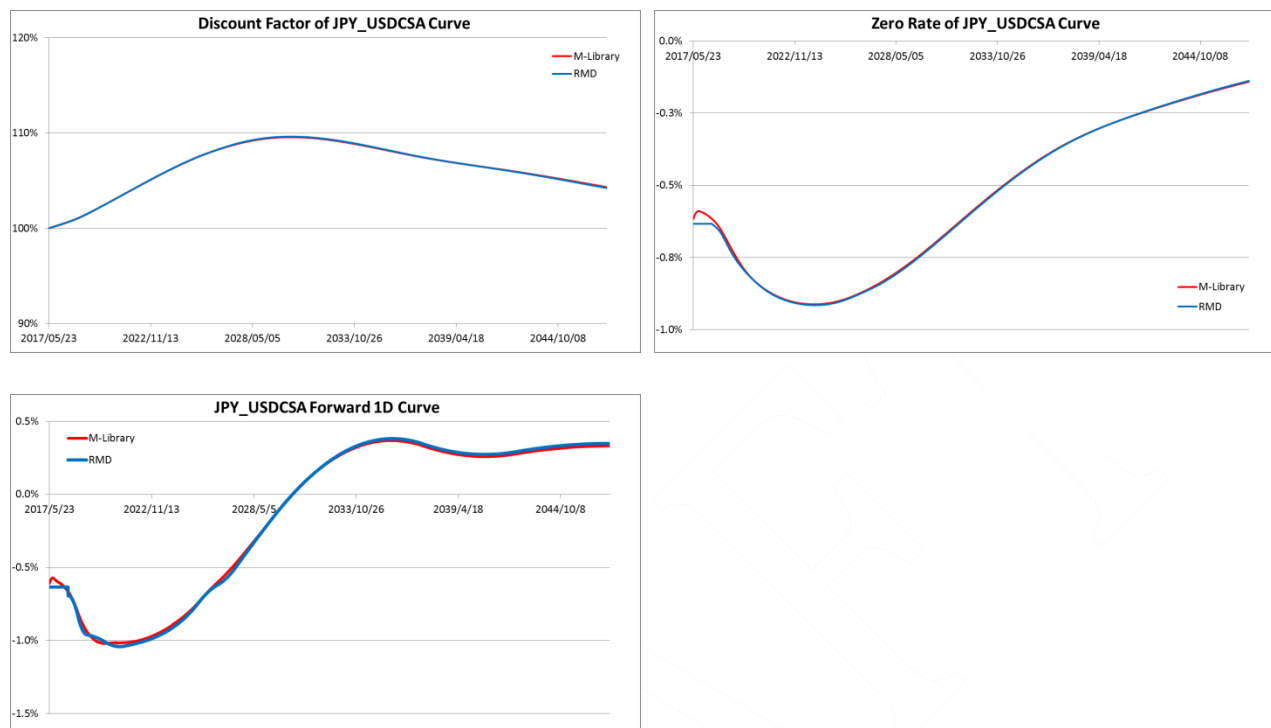


Figure 4.3.4 JPY with USD CSA collateral Curves

The M-Library curves generally match with our benchmark curves independently built in RMD library. The differences in the short end of JPY with USD CSA Collateral Curves are coming from the different extrapolation.

For the forward OIS curve, the curves constructed in RMD library look slightly smoother, with the curtain-like shape at long end disappearing. It may be because the iterative bootstrap in RMD library tried to solve the discount factors globally, whereas the M-Library tries to find the discount factors sequentially with 1 dimensional Newton Raphson for OIS curve. The iterative bootstrap may be considered as an alternative method of the curve builder.

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